

Real World Dataset: E-Commerce Sales Analytics

Imagine you work as a **Data Engineer / Business Analyst** for an online shopping company like Amazon, Flipkart, or Myntra.

Dataset: sales.csv

```
OrderID, CustomerID, CustomerName, Product, Category, Region, OrderDate, Quantity, Price
1001, C001, John Doe, Laptop, Electronics, North, 2025-01-01, 2, 50000
1002, C002, Mary Smith, Mobile, Electronics, South, 2025-01-02, 1, 25000
1003, C001, John Doe, Mouse, Accessories, North, 2025-01-03, 3, 500
1004, C003, Ravi Kumar, Keyboard, Accessories, West, 2025-01-04, 2, 1500
1005, C004, Priya Sharma, Laptop, Electronics, East, 2025-01-05, 1, 52000
1006, C002, Mary Smith, Headphones, Accessories, South, 2025-01-06, 2, 2000
```

1. Read Data

Business Use Case

Load sales data from CSV files received daily.

```
import pandas as pd
```

```
df = pd.read_csv("sales.csv")
```

```
print(df.head())
```

2. Data Understanding

Business Question

How many records do we have?

```
print(df.shape)
```

Output

```
(6, 9)
```

Business Question

What columns exist?

```
print(df.info())
```

Business Question

Summary statistics for sales?

```
print(df.describe())
```

3. Create New Columns

Business Use Case

Calculate Revenue.

Revenue = Quantity × Price

```
df["Revenue"] = df["Quantity"] * df["Price"]
```

Output

OrderID	Product	Revenue
1001	Laptop	100000
1002	Mobile	25000
1003	Mouse	1500

Using Cheat Sheet Function:

```
df = df.assign(  
    Revenue=lambda x: x.Quantity * x.Price  
)
```

4. Filtering Rows

Business Question

Show all orders above ₹50,000 revenue.

```
df[df["Revenue"] > 50000]
```

or

```
df.query("Revenue > 50000")
```

Output

OrderID	Product	Revenue
1001	Laptop	100000
1005	Laptop	52000

5. Selecting Columns

Business Question

Show only customer and revenue information.

```
df[["CustomerName", "Revenue"]]
```

6. Sorting Data

Business Question

Who generated the highest revenue?

```
df.sort_values(  
    "Revenue",  
    ascending=False  
)
```

Output

Customer	Revenue
John Doe	100000
Priya Sharma	52000

7. Group By

Business Question

Revenue by Region

```
df.groupby("Region")["Revenue"].sum()
```

Output

East	52000
North	101500
South	29000
West	3000

8. Aggregation

Business Question

Average Revenue per Region

```
df.groupby("Region").agg({  
    "Revenue":["sum","mean","max"]  
})
```

Output

Region	Sum	Mean	Max
North	101500	50750	100000

9. Top Selling Products

Business Question

Top 3 products by revenue.

```
df.nlargest(3, "Revenue")
```

Output

Product	Revenue
Laptop	100000
Laptop	52000
Mobile	25000

10. Missing Value Handling

Suppose:

```
df.loc[2,"Price"] = None
```

Find Missing Values

```
df.isnull().sum()
```

Replace Missing Values

```
df["Price"] = df["Price"].fillna(  
    df["Price"].mean()  
)
```

11. Remove Duplicate Customers

Business Question

Unique customer list for marketing campaigns.

```
customers = df.drop_duplicates(  
    subset=["CustomerID"]  
)
```

12. Date Analysis

Convert date column.

```
df["OrderDate"] = pd.to_datetime(  
    df["OrderDate"]  
)
```

Extract Year

```
df["Year"] = df["OrderDate"].dt.year
```

Extract Month

```
df["Month"] = df["OrderDate"].dt.month
```

Extract Quarter

```
df["Quarter"] = df["OrderDate"].dt.quarter
```

Business Use Case

Quarterly Revenue Reporting

```
df.groupby("Quarter")["Revenue"].sum()
```

13. Ranking Customers

Business Question

Top Customers by Revenue

```
customer_sales = (  
    df.groupby("CustomerName")  
      ["Revenue"]  
      .sum()  
      .reset_index()  
)
```

```
customer_sales["Rank"] = (  
    customer_sales["Revenue"]  
      .rank(  
          method="dense",  
          ascending=False  
      )  
)
```

Output

Customer	Revenue	Rank
John Doe	101500	1
Priya Sharma	52000	2

14. Rolling Window Analysis

Business Question

7-Day Moving Average Sales

```
daily_sales = (  
    df.groupby("OrderDate")  
      ["Revenue"]  
      .sum()  
      .reset_index()  
)  
  
daily_sales["MovingAvg"] = (  
    daily_sales["Revenue"]  
      .rolling(7)  
      .mean()  
)
```

Used heavily in:

- Retail
 - Banking
 - Stock Markets
 - Forecasting
-

15. Cumulative Revenue

Business Question

How much revenue has accumulated till date?

```
daily_sales["CumulativeRevenue"] = (  
    daily_sales["Revenue"]  
      .cumsum()  
)
```

Output

Date	Revenue	Cumulative
------	---------	------------

Jan1	100000	100000
Jan2	25000	125000
Jan3	1500	126500

16. Join Multiple Datasets

Customers

```
customers = pd.DataFrame({  
    "CustomerID":["C001","C002","C003"],  
    "City":["Mumbai","Chennai","Delhi"]  
})
```

Merge

```
result = pd.merge(  
    df,  
    customers,  
    on="CustomerID",  
    how="left"  
)
```

Business Use Case

Customer Segmentation

17. Pivot Table

Business Question

Revenue by Region and Category

```
pivot = df.pivot_table(  
    values="Revenue",  
    index="Region",  
    columns="Category",
```

```
aggfunc="sum"  
)
```

Output

Region	Accessories	Electronics
North	1500	100000
South	4000	25000

18. Melt (Unpivot)

Suppose management wants:

```
Month  Metric  Value  
pd.melt(df)
```

Useful in:

- Dashboard tools
 - Power BI
 - Tableau
-

19. String Operations

Business Question

Find customers whose names start with "J"

```
df[  
    df["CustomerName"]  
    .str.startswith("J")  
]
```

Convert names to Title Case

```
df["CustomerName"] = (  
    df["CustomerName"]  
    .str.title()  
)
```

20. Visualization

Revenue by Product

```
import matplotlib.pyplot as plt
```

```
df.groupby("Product")["Revenue"] \  
    .sum() \  
    .plot.bar()
```

```
plt.show()
```

End-to-End Business Dashboard KPIs

Using only the cheat sheet functions, a company can build:

Sales KPIs

```
total_revenue = df["Revenue"].sum()  
avg_order_value = df["Revenue"].mean()  
top_product = (  
    df.groupby("Product")  
        ["Revenue"]  
        .sum()  
        .idxmax()  
)
```

Customer KPIs

```
unique_customers = (  
    df["CustomerID"]  
    .nunique()  
)
```

Regional KPIs

```
region_sales = (  
    df.groupby("Region")  
      ["Revenue"]  
      .sum()  
)
```

Time KPIs

```
monthly_sales = (  
    df.groupby("Month")  
      ["Revenue"]  
      .sum()  
)
```

Common Industries Where These Pandas Functions Are Used

Industry	Use Case	Pandas Functions
Banking	Transaction Analysis	groupby, rolling, merge
Healthcare	Patient Analytics	query, fillna, pivot
Retail	Sales Dashboard	groupby, sort_values, plot
Manufacturing	Production Metrics	cumsum, rolling, rank
Logistics	Shipment Tracking	merge, datetime
E-commerce	Customer Segmentation	merge, groupby, pivot
Marketing	Campaign Analysis	melt, pivot, query
Finance	Stock Analysis	rolling, cumsum, rank

This single E-commerce dataset can be used to teach **80–90% of the Pandas cheat sheet functions** in a realistic Data Engineering and Business Analytics workflow.